# Functions & Transformations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Domain and range

State the domain of $f(x) = \sqrt{x - 3}$.

$x - 3 \geq 0$, so the domain is $x \geq 3$.

**2** [Foundational] • Target: Domain and range

State the range of $f(x) = x^2 + 2$.

Since $x^2 \geq 0$ for all real $x$, the range is $f(x) \geq 2$.

**3** [Foundational] • Target: Transformations: shifts

The graph of $y = x^2$ is shifted $3$ units right and $2$ units down. Write the new equation.

$y = (x - 3)^2 - 2$.

**4** [Foundational] • Target: Transformations: shifts

The graph of $y = |x|$ is shifted $4$ units left and $1$ unit up. Write the new equation.

$y = |x + 4| + 1$.

**5** [Foundational] • Target: Transformations: shifts

Describe the transformation that takes $y = x^2$ to $y = (x + 2)^2 - 5$.

A shift $2$ units left and $5$ units down.

**6** [Foundational] • Target: Transformations: reflections and stretches

The graph of $y = x^2$ is reflected over the $x$-axis and stretched vertically by a factor of $3$. Write the new equation.

$y = -3x^2$.

**7** [Foundational] • Target: Composite functions

Using the same $f$ and $g$, find $(g \circ f)(2)$.

$f(2) = 5$. $g(5) = 25 - 3 = 22$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

© ClipSAT — clipsat.org | All Rights Reserved

**8** [Intermediate] • Target: Composite functions

Using the same $f$ and $g$, find a general formula for $(f \circ g)(x)$.

$f(g(x)) = 2(x^2 - 3) + 1 = 2x^2 - 6 + 1 = 2x^2 - 5$.

**9** [Intermediate] • Target: Inverse functions

Find the inverse of $f(x) = 3x - 2$.

$y = 3x - 2 \Rightarrow x = 3y - 2 \Rightarrow y = \frac{x+2}{3}$. So $f^{-1}(x) = \frac{x+2}{3}$.

**10** [Intermediate] • Target: Inverse functions

Show that $f(x) = 2x + 6$ and $g(x) = \frac{x-6}{2}$ are inverses by computing $f(g(x))$.

$f(g(x)) = 2\left(\frac{x-6}{2}\right) + 6 = (x - 6) + 6 = x$, confirming they are inverses.

**11** [Intermediate] • Target: Piecewise functions

A function is defined by $f(x) = x + 2$ for $x < 1$, and $f(x) = 3x - 1$ for $x \geq 1$. Find $f(-2)$.

Since $-2 < 1$, use the first rule: $f(-2) = -2 + 2 = 0$.

**12** [Intermediate] • Target: Piecewise functions

Using the same piecewise function, find $f(1)$.

Since $1 \geq 1$, use the second rule: $f(1) = 3(1) - 1 = 2$.

**13** [Intermediate] • Target: Even and odd functions

Determine whether $f(x) = x^2 - 3$ is even, odd, or neither.

$f(-x) = (-x)^2 - 3 = x^2 - 3 = f(x)$, so $f$ is even.

**14** [Intermediate] • Target: Increasing, decreasing, and average rate of change

Find the average rate of change of $f(x) = x^2$ from $x = 1$ to $x = 4$.

$\frac{f(4) - f(1)}{4 - 1} = \frac{16 - 1}{3} = \frac{15}{3} = 5$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Increasing, decreasing, and average rate of change • Cross-Lesson: 3.01 Sequences & Series

State the interval on which $f(x) = (x - 2)^2$ is decreasing.

The vertex is at $x = 2$ and the parabola opens upward, so $f$ is decreasing for $x < 2$.

© ClipSAT — clipsat.org | All Rights Reserved

**16** [Challenging] • Target: Reading a function graph • Cross-Lesson: 4.01 Trigonometry

The graph of $y = f(x) = x^2 - 4$ is shown.

**a** State the zeros of $f$. $x = -2$ and $x = 2$ (where the curve crosses the $x$-axis).

**b** State the domain and range of $f$. Domain: all real numbers. Range: $y \geq -4$.

(a) $x = -2$ and $x = 2$ (where the curve crosses the $x$-axis).; (b) Domain: all real numbers. Range: $y \geq -4$.

**17** [Challenging] • Target: Word problems: functions in context • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

This question has two parts. A company's daily profit (in dollars) from selling $x$ units is modeled by $P(x) = -2x^2 + 80x - 150$.

**a** Evaluate $P(10)$ to find the profit from selling $10$ units.

$P(10) = -2(100) + 800 - 150 = -200 + 800 - 150 = 450$ dollars.

**b** Find the number of units $x$ that maximizes profit (using the vertex of the parabola), and state the maximum profit. The vertex is at $x = -\frac{80}{2(-2)} = 20$ units. $P(20) = -2(400) + 1600 - 150 = 650$ dollars — the maximum profit.

(a) $P(10) = -2(100) + 800 - 150 = -200 + 800 - 150 = 450$ dollars.; (b) The vertex is at $x = -\frac{80}{2(-2)} = 20$ units. $P(20) = -2(400) + 1600 - 150 = 650$ dollars — the maximum profit.

**18** [Challenging] • Target: Word problems: functions in context • Cross-Lesson: 6.01 Limits & Derivatives

This question has two parts. The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9}{5}C + 32$. Meanwhile, the temperature in a city $t$ hours after 6 a.m. is modeled by $C(t) = -0.5t^2 + 6t + 10$ (in degrees Celsius), for $0 \leq t \leq 12$.

**a** Find $C(4)$, the Celsius temperature $4$ hours after 6 a.m.

$C(4) = -0.5(16) + 24 + 10 = -8 + 24 + 10 = 26°\text{C}$.

**b** Find $(F \circ C)(4)$, the Fahrenheit temperature at that time.

$(F \circ C)(4) = \frac{9}{5}(26) + 32 = 46.8 + 32 = 78.8°\text{F}$.

(a) $C(4) = -0.5(16) + 24 + 10 = -8 + 24 + 10 = 26°\text{C}$.; (b) $(F \circ C)(4) = \frac{9}{5}(26) + 32 = 46.8 + 32 = 78.8$ °F.

**19** [Challenging] • Target: MCQ — Piecewise functions • Cross-Lesson: 7.01 Integration & Applications

Find the domain of the function $h(x) = \frac{1}{\sqrt{x-4}}$.

Step 1: Identify domain restrictions: Two conditions apply: the expression under the square root must be non-negative ($x - 4 \geq 0$), and the denominator cannot be zero ($\sqrt{x-4} \neq 0$). Step 2: Combine restrictions: Combining both conditions requires the radicand to be strictly positive: $x - 4 > 0 \Rightarrow x > 4$. In interval notation: **$(4, \infty)$**.

© ClipSAT — clipsat.org | All Rights Reserved

**20** [Challenging] • Target: MCQ — Even and odd functions • Cross-Lesson: 8.01 Exponentials & Logarithms

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{2x-5}{3}$.

Step 1: Set $y = f(x)$ and swap variables: $y = \frac{2x-5}{3} \Rightarrow x = \frac{2y-5}{3}$. Step 2: Solve for $y$: $3x = 2y - 5 \Rightarrow 2y = 3x + 5 \Rightarrow y = \frac{3x+5}{2}$. Therefore: $\mathbf{f^{-1}(x) = \frac{3x+5}{2}}$.

---

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: MCQ — Domain and range

The graph shows $f(x)$ defined only where plotted. State its domain.

**a.** $[-3, 3]$ ✓

**b.** All real numbers

**c.** $[-4, 4]$

**d.** $[0, 3]$

The curve is only plotted for $x$ between $-3$ and $3$.

**22** [Foundational] • Target: MCQ — Transformations: shifts

The graph of $y = |x|$ is shifted 2 units left and 4 units up. Write the new equation.

**a.** $y = |x + 4| + 2$

**b.** $y = |x + 2| + 4$ ✓

**c.** $y = |x + 2| - 4$

**d.** $y = |x - 2| + 4$

Left $2$: replace $x$ with $x + 2$; up $4$: add $4$.

**23** [Foundational] • Target: MCQ — Transformations: shifts

The graph of $y = \sqrt{x}$ is shifted 6 units right. Write the new equation.

**a.** $y = \sqrt{x + 6}$

**b.** $y = \sqrt{x} - 6$

**c.** $y = \sqrt{x} + 6$

**d.** $y = \sqrt{x - 6}$ ✓

Right $6$: replace $x$ with $x - 6$.

© ClipSAT — clipsat.org | All Rights Reserved

**24** [Foundational] • Target: MCQ — Transformations: reflections and stretches

The graph of $y = \sqrt{x}$ is reflected over the $y$-axis. Write the new equation.

**a.** $y = -\sqrt{-x}$

**b.** $y = -\sqrt{x}$

**c.** $y = \sqrt{-x}$ ✓

**d.** $y = \sqrt{x}$

Reflecting over the $y$-axis replaces $x$ with $-x$: $y = \sqrt{-x}$.

**25** [Foundational] • Target: MCQ — Even and odd functions

The graph shown is symmetric about the $y$-axis. Is the function even, odd, or neither?

**a.** Odd

**b.** Neither

**c.** Cannot be determined

**d.** **Even** ✓

Symmetry about the $y$-axis is exactly the definition of an even function.

**26** [Foundational] • Target: MCQ — Reading a function graph

The graph shows $y = f(x) = x^2 - 9$. Find $f(3)$ by reading the graph.

**a.** $-9$

**b.** $6$

**c.** $9$

**d.** $0$ ✓

The curve crosses the $x$-axis at $x = 3$, so $f(3) = 0$.

**27** [Foundational] • Target: MCQ — Reading a function graph

Using the same graph, find the values of $x$ where $f(x) = 0$ (the $x$-intercepts).

**a.** $x = 0$ and $x = 9$

**b.** $x = -9$ and $x = 9$

**c.** $x = 3$ only

**d.** $x = -3$ **and** $x = 3$ ✓

The curve crosses the $x$-axis at $x = -3$ and $x = 3$.

**28** [Foundational] • Target: MCQ — Reading a function graph

Using the same graph, find the $y$-intercept.

**a.** $-3$

**b.** $9$

**c.** $0$

**d.** $-9$ ✓

At $x = 0$, the curve crosses the $y$-axis at $y = -9$.

© ClipSAT — clipsat.org | All Rights Reserved

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: MCQ — Word problems: functions in context

A company's daily profit (in dollars) from selling $x$ units is modeled by $P(x) = -2x^2 + 80x - 300$. Find the profit when 15 units are sold.

**a.** 450 **dollars** ✓

**b.** 150 dollars

**c.** 600 dollars

**d.** 900 dollars

$P(15) = -2(225) + 1{,}200 - 300 = 450$ dollars.

**30** [Intermediate] • Target: MCQ — Word problems: functions in context

The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9C}{5} + 32$. Find $F(20)$.

**a.** 72°F

**b.** 68°**F** ✓

**c.** 36°F

**d.** 52°F

$F(20) = \frac{9(20)}{5} + 32 = 36 + 32 = 68$.

**31** [Intermediate] • Target: MCQ — Word problems: functions in context

A ball's height (in meters) $t$ seconds after being thrown is $h(t) = -5t^2 + 20t + 2$. Find the height at $t = 2$ seconds.

**a.** 2

**b.** 18

**c.** 22 ✓

**d.** 42

$h(2) = -5(4) + 40 + 2 = 22$ meters.

**32** [Intermediate] • Target: Word problems: functions in context

The graph of $y = x^2$ is shifted 1 unit left and 4 units down. Write the new equation.

**a.** $y = (x - 1)^2 + 4$

**b.** $y = (x + 1)^2 + 4$

**c.** $y = (x + 1)^2 - 4$ ✓

**d.** $y = (x - 1)^2 - 4$

$y = (x + 1)^2 - 4$.

© ClipSAT — clipsat.org | All Rights Reserved

**33** [Intermediate] • Target: MCQ — Evaluating functions

The graph of $y = |x|$ is shifted 7 units left and 3 units down. Write the new equation.

**a.** $y = |x + 7| + 3$

**b.** $y = |x - 7| - 3$

**c.** $y = |x + 7| - 3$ ✓

**d.** $y = |x - 7| + 3$

$y = |x + 7| - 3$.

**34** [Intermediate] • Target: MCQ — Domain and range

■■ ■■■■■ ■■■■■ ■■■■■■■ ■■■■■■■ $f(x) = x^2 - 3x$ ■■■ ■■■■■■■ $[1, 4]$■ Average rate of change of $x^2 - 3x$ on $[1, 4]$?

**a.** **2** ✓

**b.** 3

**c.** 4

**d.** 5

$f(1) = 1 - 3 = -2$■ $f(4) = 16 - 12 = 4$. ■■■■■ ■■■■■ ■■■■■■■ $= \frac{4 - (-2)}{4 - 1} = \frac{6}{3} = 2$.

**35** [Intermediate] • Target: MCQ — Transformations: shifts

■■■■■■ ■■■■■■ ■■■■■ ■■■■■■ ■■■■■■■■ ■■■■■■■ $f(x) = x^2 - 5x + k$. ■■■■ ■■■■ ■■■■■■■■ ■■■■■ ■■■■■■■ $x$ ■■■■ $x = 2$■ ■■■■ ■■ ■■■■■ $k$■ The graph below represents the function $f(x) = x^2 - 5x + k$. If the curve intersects the $x$-axis at $x = 2$, what is the value of $k$?

**a.** **–6** ✓

**b.** 6

**c.** –4

**d.** 4

■■■■ ■■ ■■■■■■■■ ■■■■■ ■■■■■■■ $x$ ■■■■ $x = 2$■ ■■■■■ ■■■■■ ■■ $f(2) = 0$. ■■■■■ $x = 2$ ■■ ■■■■■■■ ■■■■■■■: $f(2) = (2)^2 - 5(2) + k = 0$ $4 - 10 + k = 0$ $-6 + k = 0$ $k = 6$ ■■■■■ ■■■■■ $k$ ■■ 6. --- Since the curve intersects the $x$-axis at $x = 2$, this means that $f(2) = 0$. We substitute $x = 2$ into the function's equation: $f(2) = (2)^2 - 5(2) + k = 0$ $4 - 10 + k = 0$ $-6 + k = 0$ $k = 6$ Therefore, the value of $k$ is 6.

© ClipSAT — clipsat.org | All Rights Reserved

**36** [Intermediate] • Target: MCQ — Transformations: reflections and stretches

■■ ■■■■■ ■■■■■■■ $f(x) = \sqrt[3]{x-5}$■ Domain of cube root function $\sqrt[3]{x-5}$?

**a.** $\mathbb{R}$ ✓

**b.** $[5, \infty)$

**c.** $(5, \infty)$

**d.** $\mathbb{R} \setminus \{5\}$

■■■■■ ■■ ■■■■■■■ ■■■■■■■ (■■■■■■■■■) ■■■■■■ ■■■■■■ ■■■■■■■■■ ■■■■■■■■■ ■■■■■■■■ ■■■■■■■■■ ■■■■■■■: $\mathbb{R}$.

**37** [Intermediate] • Target: MCQ — Composite functions

■■ ■■■■■■■ ■■■■■■■■ ■■ ■■■■■ ■■■■■■ ■■■■■■■■ ■■■■ ■■■■■ ■■■■■■■ Which is an odd function symmetric about origin?

**a.** $f(x) = x^3 - x$ ✓

**b.** $f(x) = x^2 - 1$

**c.** $f(x) = x^4 + 3$

**d.** $f(x) = |x|$

$f(-x) = (-x)^3 - (-x) = -x^3 + x = -(x^3 - x) = -f(x)$■ ■■■■ ■■■■■ ■■■■■■.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: MCQ — Inverse functions • Cross-Lesson: 6.01 Limits & Derivatives

■■ ■■■■■■■ ■■■■■■■■ ■■ ■■■■■ ■■■■■■ ■■■■■■■■ ■■■■ ■■■■■■■ $y$■ Which is an even function symmetric about y-axis?

**a.** $f(x) = x^2 + 4$ ✓

**b.** $f(x) = x^3$

**c.** $f(x) = x^3 + x$

**d.** $f(x) = 2x + 1$

$f(-x) = (-x)^2 + 4 = x^2 + 4 = f(x)$■ ■■■■ ■■■■ ■■■■■ ■■■■■■.

© ClipSAT — clipsat.org | All Rights Reserved

**39** [Challenging] • Target: MCQ — Piecewise functions • Cross-Lesson: 7.01 Integration & Applications

[illegible] $f(x)$ [illegible]. [illegible] [illegible] $\lim_{x\to 2} f(x)$[illegible] The graph of the function $f(x)$ is shown below. What is the value of $\lim_{x\to 2} f(x)$?

**a.** 0

**b.** 1

**c.** **2** ✓

**d.** 4

[illegible] [illegible] [illegible] $f(x)$ [illegible] [illegible] $x$ [illegible] 2 ($\lim_{x\to 2} f(x)$) [illegible] [illegible] [illegible] $y$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $x = 2$ [illegible] [illegible] [illegible] ([illegible] [illegible]). [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $x$ [illegible] 2 [illegible] [illegible] ([illegible] [illegible] [illegible] 2)[illegible] [illegible] $f(x)$ [illegible] [illegible] 2. [illegible] [illegible] $x$ [illegible] 2 [illegible] [illegible] ([illegible] [illegible] [illegible] 2)[illegible] [illegible] $f(x)$ [illegible] [illegible] [illegible] 2. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

**41** [Challenging] • Target: MCQ — Increasing, decreasing, and average rate of change • Cross-Lesson: 9.01 Statistics & Probability

[redacted] $f(x) = -2x^2 + 8x - 5$ [redacted]. [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] The function $f(x) = -2x^2 + 8x - 5$ represents the trajectory of a projectile. What is the maximum height the projectile reaches?

**a.** 4 [redacted] 4 units

**b.** **3 [redacted] 3 units ✓**

**c.** 5 [redacted] 5 units

**d.** 11 [redacted] 11 units

[redacted] [redacted] [redacted] [redacted] $f(x) = ax^2 + bx + c$[redacted] [redacted] $a = -2$[redacted] $b = 8$[redacted] [redacted] $c = -5$. [redacted] [redacted] $x$, where $a = -2$, $b = 8$, and $c = -5$. Since $x$. The x-coordinate of the vertex is given by the formula $x = -\frac{b}{2a}$. [redacted] [redacted] [redacted] [redacted]: $x = -\frac{8}{2(-2)} = -\frac{8}{-4} = 2$. Substituting the values, we get: $x = -\frac{8}{2(-2)} = -\frac{8}{-4} = 2$. [redacted] [redacted] [redacted] ([redacted] [redacted]) [redacted] [redacted] $x = 2$. [redacted] [redacted] $x = 2$ [redacted] [redacted] $f(x)$ [redacted] [redacted] [redacted]. The maximum value o.

**42** [Challenging] • Target: MCQ — Operations on functions • Cross-Lesson: 1.01 Algebra & Equations

[redacted] [redacted] [redacted] [redacted] $f(x)$ [redacted] [redacted] [redacted] $x = 2$. [redacted] [redacted] [redacted] $f(x)$ [redacted] [redacted] $x$ [redacted] 2[redacted] The graph of the function $f(x)$ is shown. What is the limit of $f(x)$ as $x$ approaches 2?

**a.** $-1$

**b.** 2 ✓

**c.** 0

**d.** 3

[redacted] [redacted] [redacted] [redacted] [redacted] $f(x)$ [redacted] [redacted] $x$ [redacted] 2 [redacted] [redacted] [redacted]. [redacted] [redacted] [redacted] [redacted] [redacted] $x$ [redacted] 2 [redacted] [redacted] ([redacted] [redacted] [redacted] 2)[redacted] [redacted] $f(x)$ [redacted] [redacted] 2. [redacted] [redacted] $x$ [redacted] 2 [redacted] [redacted] ([redacted] [redacted] [redacted] 2)[redacted] [redacted] $f(x)$ [redacted] [redacted] [redacted] 2. [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] $f(x)$ [redacted] [redacted] $x$ [redacted] 2 [redacted] 2. We look for the value that $f(x)$ approaches as $x$ approaches 2 from both sides. Looking at the graph, as $x$ approaches 2 from the left (values less tha.

© ClipSAT — clipsat.org | All Rights Reserved

**43** [Challenging] • Target: MCQ — Reading a function graph • Cross-Lesson: 3.01 Sequences & Series

■■■■■■■ ■■■■ ■■■■■■■ $f(x) = x^2 - 4x + 3$■ ■■ ■■ ■■■■■ ■■■■■■■■■ ■■ ■■■■■■■■ ■■■■■■■■■

Given the function $f(x) = x^2 - 4x + 3$, what are its x-intercepts?

**a.** $(1, 0)$ ■ $(3, 0)$ ✓

**b.** $(-1, 0)$ ■ $(-3, 0)$

**c.** $(1, 0)$ ■ $(-3, 0)$

**d.** $(-1, 0)$ ■ $(3, 0)$

■■■■■■■ ■■■■■ ■■■■■■■■■ ■■ ■■■■■■■ ■■■■■■■■■ ■■■■■ $f(x) = 0$. ■■■■■■ $x^2 - 4x + 3 = 0$. ■■■■■ ■■■■■■ ■■■■ ■■■■■■■■■■ ■■■■■■■■■■■ ■■■■ $(x - 1)(x - 3) = 0$. ■■■■ ■■■■■■■■ ■■■■■■: $x = 1$ ■ $x = 3$. ■■■■■ ■■■■■ ■■■■■■■■■ ■■ ■■■■■■■ ■■■■■■■ ■■ $(1, 0)$ ■ $(3, 0)$. To find the x-intercepts, we set $f(x) = 0$. Therefore, $x^2 - 4x + 3 = 0$. This quadratic equation can be factored as $(x - 1)(x - 3) = 0$. This gives us two roots: $x = 1$ and $x = 3$. Thus, the x-intercepts are $(1, 0)$ and $(3, 0)$.

**44** [Challenging] • Target: MCQ — Word problems: functions in context • Cross-Lesson: 4.01 Trigonometry

■■■■ ■■■■■ $f(x) = x^2 - 2$ ■ $g(x) = \sqrt{x}$ ■ ■■■■ ■■ ■■■■■ $(f \circ g)(x)$ ■■■■■■■■■ Formula and domain of $(f \circ g)(x)$?

**a.** $x - 2$ ■■■■■■■■ $[0, \infty)$ ✓

**b.** $x - 2$ ■■■■■■■■ $\mathbb{R}$

**c.** $\sqrt{x^2 - 2}$ ■■■■■■■■ $[0, \infty)$

**d.** $x^2 - 2$ ■■■■■■■■ $\mathbb{R}$

$(f \circ g)(x) = (\sqrt{x})^2 - 2 = x - 2$. ■■■■ ■■ ■■■■■ $g(x)$ ■■ $[0, \infty)$■ ■■■■ ■■■■■ ■■■■■■■■ ■■ $[0, \infty)$ .

**45** [Challenging] • Target: Evaluating functions • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

■■ ■■■■ ■■■■■■■ $f(x) = x^2 + 3$■ Range of $f(x) = x^2 + 3$?

**a.** $[3, \infty)$ ✓

**b.** $(3, \infty)$

**c.** $[0, \infty)$

**d.** $\mathbb{R}$

■■■■ ■■ $x^2 \geq 0$ ■■■■■■■■ ■■■■ $x^2 + 3 \geq 3$■ ■■■■■■■ ■■ $[3, \infty)$.

© ClipSAT — clipsat.org | All Rights Reserved